Example 1: Sweet Apple Acres

Applejack has come to us: "Between the Red Delicious row, the Honeycrisp row, and that new Zap Apple row Granny Smith insisted on plantin', I reckon one of 'em oughta be doin' better than the others. Can you tell me if they're really any different, or am I just imaginin' things?"

- Glimpse of data:

```
# A tibble: 3 × 2
  row               bushels
  <chr>               <dbl>
1 Red Delicious Row    42.2
2 Red Delicious Row    45.3
3 Red Delicious Row    42.1
```

- What are the levels in our factor (`row`)?

```
apple_harvest |> n_pct(row)
```

```
               row    n (pct)
    Honeycrisp Row 18 (33.3%)
 Red Delicious Row 18 (33.3%)
     Zap Apple Row 18 (33.3%)
```

- What are the summary statistics by level of `row`?

```
apple_harvest |> group_by(row) |> mean_median(bushels)
```

```
# A tibble: 3 × 4
  row               variable mean_sd    median_iqr
  <chr>             <chr>    <chr>      <chr>
1 Honeycrisp Row    bushels  41.6 (8.0) 41.0 (6.9)
2 Red Delicious Row bushels  40.6 (3.8) 42.0 (5.1)
3 Zap Apple Row     bushels  37.5 (4.4) 38.5 (5.1)
```

- Constructing the ANOVA table,

```
apple_harvest |> anovas(outcome = bushels,
                        grouping = row)
```

```
One-way ANOVA

Source     df         SS       MS      F  p-value
──────────────────────────────────────────────────
row         2   160.9693  80.4846  2.468    0.095
Residuals  51  1663.4383  32.6164
──────────────────────────────────────────────────


Hypotheses (row):
H₀: All group means are equal (no effect of row)
H₁: At least one group mean differs (effect of row exists)
Test statistic: F(2, 51) = 2.468
p-value: 0.095
Conclusion: Fail to reject the null hypothesis (p = 0.095 ≥ α = 0.050)
```

- **Hypotheses:**

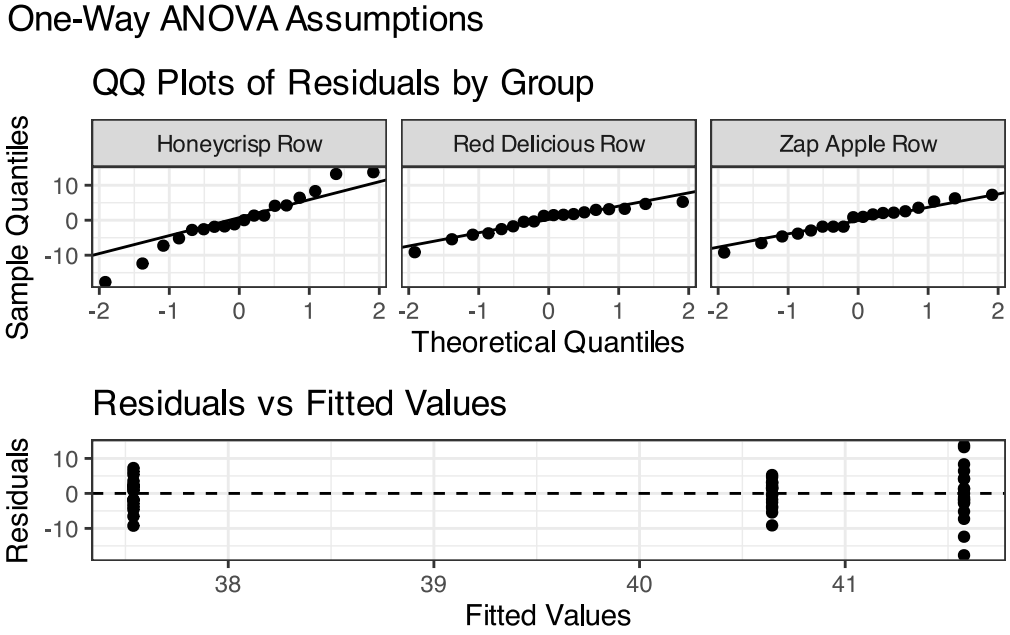
- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

- Checking the normality of our Sweet Apple Acres dataset,

```
apple_harvest |> anova_check(outcome = bushels,
                             grouping = row)
```



- Applying the Kruskal-Wallis,

```
apple_harvest |> kruskal_wallis(outcome = bushels,
                                grouping = row)
```

```
Kruskal-Wallis test

M[Honeycrisp Row] = 41 (n = 18)
M[Red Delicious Row] = 42 (n = 18)
M[Zap Apple Row] = 38.45 (n = 18)

Hypotheses:
H₀: All group distributions are equal (no effect of row)
H₁: At least one group distribution differs in location (effect of row exists)
Test statistic: H(2) = 5.953
p-value: 0.051
Conclusion: Fail to reject the null hypothesis (p = 0.051 ≥ α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

Example 2: Rarity's Boutique

- Rarity flits between racks of fabric: "Darling, I need your professional opinion. Between my Canterlot Couture line, my everyday Ponyville Casual line, and my absolutely *dazzling* Gala Elegance line, I use wildly different amounts of gemstones. Is the difference really as dramatic as I think?"

```
# A tibble: 3 × 2
  collection        gem_count
  <chr>                 <dbl>
1 Canterlot Couture        59
2 Canterlot Couture        57
3 Canterlot Couture        64
```

- We will use the `boutique_gems` dataset to answer the following question:
  - Does the number of gems used per dress differ across the three collections?
- What are the levels in our factor (`collection`)?

```
boutique_gems |> n_pct(collection)
```

```
        collection    n (pct)
 Canterlot Couture 15 (33.3%)
     Gala Elegance 15 (33.3%)
  Ponyville Casual 15 (33.3%)
```

- What are the summary statistics by level of `collection`?

```
boutique_gems |> group_by(collection) |> mean_median(gem_count)
```

```
# A tibble: 3 × 4
  collection        variable  mean_sd     median_iqr
  <chr>             <chr>     <chr>       <chr>
1 Canterlot Couture gem_count 55.9 (8.4)  57.0 (8.0)
2 Gala Elegance     gem_count 124.3 (7.6) 124.0 (8.5)
3 Ponyville Casual  gem_count 87.9 (6.4)  87.0 (9.0)
```

- Constructing the ANOVA table,

```
boutique_gems |> anovas(outcome = gem_count,
                        grouping = collection)
```

```
One-way ANOVA

Source      df          SS          MS        F  p-value
──────────────────────────────────────────────────────────
collection   2  35140.5778  17570.2889  311.110  < 0.001
Residuals   42   2372.0000     56.4762
──────────────────────────────────────────────────────────


Hypotheses (collection):
H₀: All group means are equal (no effect of collection)
H₁: At least one group mean differs (effect of collection exists)
Test statistic: F(2, 42) = 311.11
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**

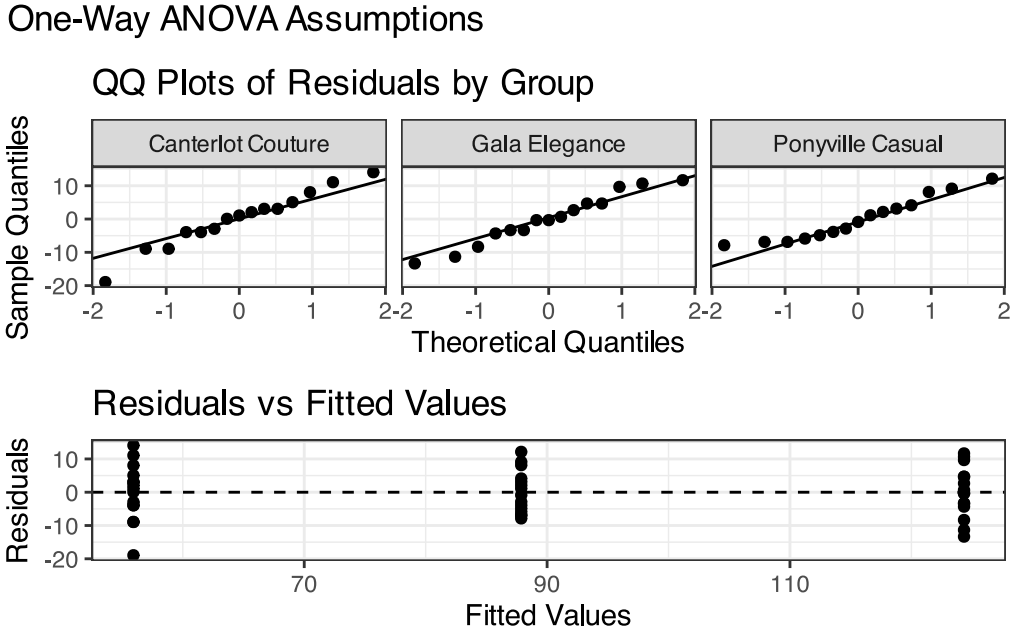
- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

- Checking the normality of Rarity's dataset,

```
boutique_gems |> anova_check(outcome = gem_count,
                             grouping = collection)
```



- Applying the Kruskal-Wallis,

```
boutique_gems |> kruskal_wallis(outcome = gem_count,
                                grouping = collection)
```

```
Kruskal-Wallis test

M[Canterlot Couture] = 57 (n = 15)
M[Gala Elegance] = 124 (n = 15)
M[Ponyville Casual] = 87 (n = 15)

Hypotheses:
H₀: All group distributions are equal (no effect of collection)
H₁: At least one group distribution differs in location (effect of collection
exists)
Test statistic: H(2) = 39.148
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

Example 3: Ponyville Diner

- Sunny Skies wipes down the counter after the lunch rush: "Feels like Caramel always walks away with better tips than me and Lyra put together. Before I start complainin' to the manager, can you check if there's actually a difference in how much we're bringing in?"

```
# A tibble: 3 × 2
  server      tip_bits
  <chr>          <dbl>
1 Sunny Skies     4.08
2 Sunny Skies     6.12
3 Sunny Skies     0.12
```

- We will use the `ponyville_diner_tips` dataset to answer the following question:
  - Does tip amount differ across the three servers?
- What are the levels in our factor (`server`)?

```
ponyville_diner_tips |> n_pct(server)
```

```
            server    n (pct)
           Caramel 22 (33.3%)
 Lyra Heartstrings 22 (33.3%)
       Sunny Skies 22 (33.3%)
```

- What are the summary statistics by level of `server`?

```
ponyville_diner_tips |> group_by(server) |> mean_median(tip_bits)
```

```
# A tibble: 3 × 4
  server            variable mean_sd   median_iqr
  <chr>             <chr>    <chr>     <chr>
1 Caramel           tip_bits 6.7 (4.4) 6.0 (5.9)
2 Lyra Heartstrings tip_bits 4.4 (4.7) 2.9 (2.9)
3 Sunny Skies       tip_bits 7.4 (8.9) 4.3 (8.5)
```

- Constructing the ANOVA table,

```
ponyville_diner_tips |> anovas(outcome = tip_bits,
                       grouping = server)
```

```
One-way ANOVA

Source     df         SS       MS      F  p-value
──────────────────────────────────────────────────
server      2   109.2923  54.6461  1.356    0.265
Residuals  63  2539.3768  40.3076
──────────────────────────────────────────────────


Hypotheses (server):
H₀: All group means are equal (no effect of server)
H₁: At least one group mean differs (effect of server exists)
Test statistic: F(2, 63) = 1.356
p-value: 0.265
Conclusion: Fail to reject the null hypothesis (p = 0.265 ≥ α = 0.050)
```

- **Hypotheses:**

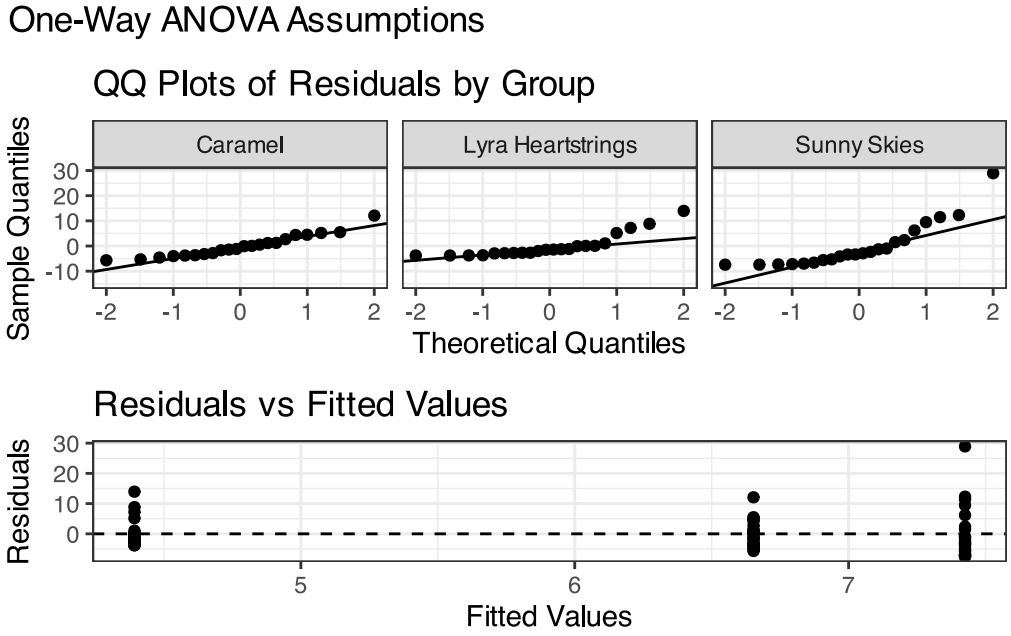
- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

- Checking the normality of the Ponyville Diner dataset,

```
ponyville_diner_tips |> anova_check(outcome = tip_bits,
                                    grouping = server)
```



- Applying the Kruskal-Wallis,

```
ponyville_diner_tips |> kruskal_wallis(outcome = tip_bits,
                       grouping = server)
```

```
Kruskal-Wallis test

M[Caramel] = 6.03 (n = 22)
M[Lyra Heartstrings] = 2.94 (n = 22)
M[Sunny Skies] = 4.305 (n = 22)

Hypotheses:
H₀: All group distributions are equal (no effect of server)
H₁: At least one group distribution differs in location (effect of server exists)
Test statistic: H(2) = 4.235
p-value: 0.120
Conclusion: Fail to reject the null hypothesis (p = 0.120 ≥ α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 1: Wonderbolts Academy

- Fleetfoot drops a stack of stopwatch printouts on the table: "Command wants to know if lap times actually differ across our four training squads: Thunderhead, Stormwing, Skybolt, and Downburst."

```
# A tibble: 3 × 2
  squad       lap_seconds
  <chr>             <dbl>
1 Thunderhead        60.4
2 Thunderhead        59.6
3 Thunderhead        62
```

- We will use the `wonderbolts_laptimes` dataset to answer the following question:
  - Do lap times differ across the four training squads?
- What are the levels in our factor (`squad`)?

```
wonderbolts_laptimes |> n_pct(squad)
```

```
       squad    n (pct)
   Downburst 20 (25.0%)
     Skybolt 20 (25.0%)
   Stormwing 20 (25.0%)
 Thunderhead 20 (25.0%)
```

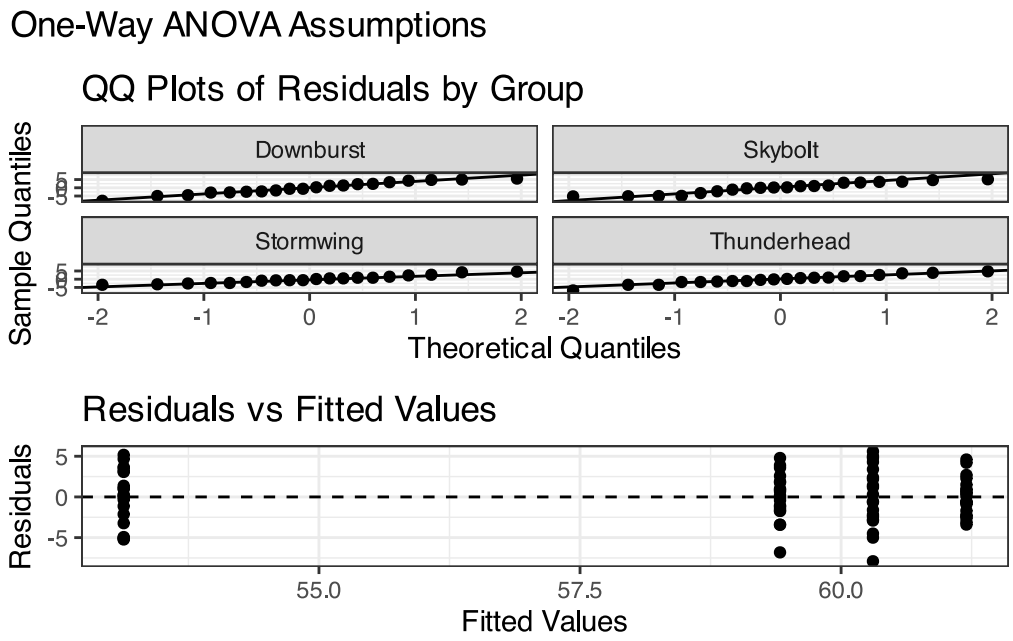
- What are the summary statistics by level of `squad`?

```
wonderbolts_laptimes |> group_by(squad) |> mean_median(lap_seconds)
```

```
# A tibble: 4 × 4
  squad       variable    mean_sd    median_iqr
  <chr>       <chr>       <chr>      <chr>
1 Downburst   lap_seconds 60.3 (3.7) 60.2 (5.2)
2 Skybolt     lap_seconds 53.1 (3.4) 53.3 (5.5)
3 Stormwing   lap_seconds 61.2 (2.3) 61.0 (2.9)
4 Thunderhead lap_seconds 59.4 (2.8) 59.5 (3.3)
```

- Checking the normality,

```
wonderbolts_laptimes |> anova_check(outcome = lap_seconds,
                                    grouping = squad)
```



- Constructing the ANOVA table,

```
wonderbolts_laptimes |> anovas(outcome = lap_seconds,
                               grouping = squad)
```

```
One-way ANOVA

Source     df        SS        MS       F  p-value
──────────────────────────────────────────────────
squad       3  804.4305  268.1435  28.098  < 0.001
Residuals  76  725.2770    9.5431
──────────────────────────────────────────────────


Hypotheses (squad):
H₀: All group means are equal (no effect of squad)
H₁: At least one group mean differs (effect of squad exists)
Test statistic: F(3, 76) = 28.098
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- Applying the Kruskal-Wallis,

```
wonderbolts_laptimes |> kruskal_wallis(outcome = lap_seconds,
                                       grouping = squad)
```

```
Kruskal-Wallis test

M[Downburst] = 60.15 (n = 20)
M[Skybolt] = 53.35 (n = 20)
M[Stormwing] = 60.95 (n = 20)
M[Thunderhead] = 59.45 (n = 20)

Hypotheses:
H₀: All group distributions are equal (no effect of squad)
H₁: At least one group distribution differs in location (effect of squad exists)
Test statistic: H(3) = 38.039
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- After choosing the appropriate method,
- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 2: Klugetown Bazaar

- A weary trader waves you over between haggling matches: "Rustbolt, Glimmering, and Junker each swear their stall pulls in the best prices. I've got a season's worth of sale records — do the three stalls actually sell for different amounts, or is everyone just talkin' up their own wares?"

```
# A tibble: 3 × 2
  stall             sale_price_bits
  <chr>                       <dbl>
1 Rustbolt's Relics            21.2
2 Rustbolt's Relics            18.9
3 Rustbolt's Relics            26.8
```

- We will use the `klugetown_bazaar` dataset to answer the following question:
  - Does sale price differ across the three market stalls?
- What are the levels in our factor (`stall`)?

```
klugetown_bazaar |> n_pct(stall)
```

```
             stall    n (pct)
  Glimmering Wares 20 (33.3%)
 Junker's Junction 20 (33.3%)
 Rustbolt's Relics 20 (33.3%)
```

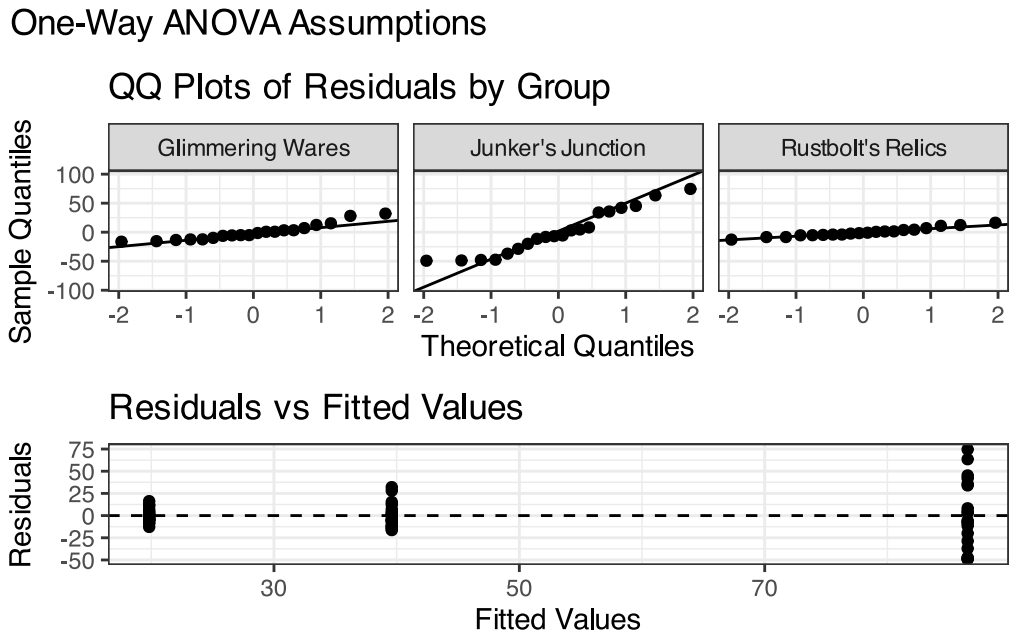
- What are the summary statistics by level of `stall`?

```
klugetown_bazaar |> group_by(stall) |> mean_median(sale_price_bits)
```

```
# A tibble: 3 × 4
  stall             variable        mean_sd     median_iqr
  <chr>             <chr>           <chr>       <chr>
1 Glimmering Wares  sale_price_bits 39.6 (13.6) 36.4 (14.7)
2 Junker's Junction sale_price_bits 86.5 (38.5) 80.3 (65.1)
3 Rustbolt's Relics sale_price_bits 19.8 (7.4)  18.6 (8.7)
```

- Checking the normality,

```
klugetown_bazaar |> anova_check(outcome = sale_price_bits,
                                grouping = stall)
```



- Constructing the ANOVA table,

```
klugetown_bazaar |> anovas(outcome = sale_price_bits,
                           grouping = stall)
```

```
One-way ANOVA

Source     df          SS          MS       F  p-value
──────────────────────────────────────────────────────
stall       2  46965.2768  23482.6384  40.982  < 0.001
Residuals  57  32661.2297    573.0040
──────────────────────────────────────────────────────


Hypotheses (stall):
H₀: All group means are equal (no effect of stall)
H₁: At least one group mean differs (effect of stall exists)
Test statistic: F(2, 57) = 40.982
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- Applying the Kruskal-Wallis,

```
klugetown_bazaar |> kruskal_wallis(outcome = sale_price_bits,
                                   grouping = stall)
```

```
Kruskal-Wallis test

M[Glimmering Wares] = 36.36 (n = 20)
M[Junker's Junction] = 80.29 (n = 20)
M[Rustbolt's Relics] = 18.565 (n = 20)

Hypotheses:
H₀: All group distributions are equal (no effect of stall)
H₁: At least one group distribution differs in location (effect of stall exists)
Test statistic: H(2) = 43.348
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- After choosing the appropriate method,
- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 3: Baltimare Harbor

- The harbormaster spreads a stack of logbooks across the table: "Four dock crews unload every ship that comes through: Anchor, Tidewater, Saltmarsh, and Driftwood. Some of the sailors have been grumbling that one crew is dragging everyone's schedule behind. Before I say anything to the crews, I want to know if unloading time actually differs across them."

```
# A tibble: 3 × 2
  crew        unload_minutes
  <chr>                <dbl>
1 Anchor Crew           1.3
2 Anchor Crew           2.19
3 Anchor Crew           5.13
```

- We will use the `baltimare_harbor` dataset to answer the following question:
  - Does unloading time differ across the four dock crews?
- What are the levels in our factor (`crew`)?

```
baltimare_harbor |> n_pct(crew)
```

```
           crew    n (pct)
    Anchor Crew 20 (25.0%)
 Driftwood Crew 20 (25.0%)
 Saltmarsh Crew 20 (25.0%)
 Tidewater Crew 20 (25.0%)
```

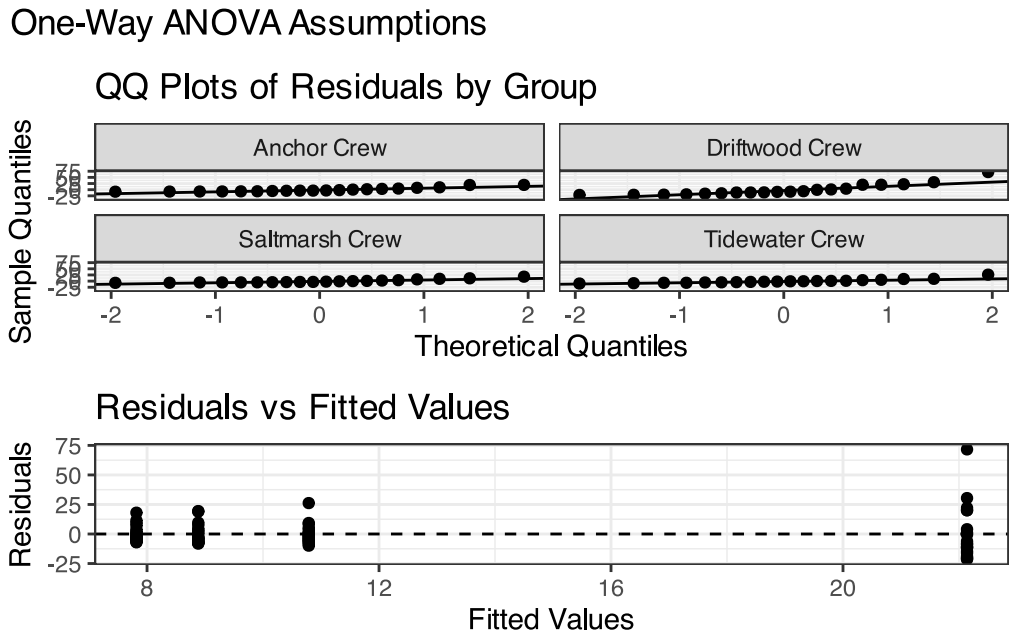
- What are the summary statistics by level of `crew`?

```
baltimare_harbor |> group_by(crew) |> mean_median(unload_minutes)
```

```
# A tibble: 4 × 4
  crew           variable       mean_sd     median_iqr
  <chr>          <chr>          <chr>       <chr>
1 Anchor Crew    unload_minutes 8.9 (8.4)   5.4 (10.0)
2 Driftwood Crew unload_minutes 22.1 (23.0) 13.7 (23.1)
3 Saltmarsh Crew unload_minutes 7.8 (6.8)   5.1 (7.6)
4 Tidewater Crew unload_minutes 10.8 (8.1)  9.5 (7.0)
```

- Checking the normality,

```
baltimare_harbor |> anova_check(outcome = unload_minutes,
                                grouping = crew)
```



- Constructing the ANOVA table,

```
baltimare_harbor |> anovas(outcome = unload_minutes,
                           grouping = crew)
```

```
One-way ANOVA

Source     df          SS        MS      F  p-value
──────────────────────────────────────────────────────
crew        3   2616.1986  872.0662  4.906    0.004
Residuals  76  13509.4136  177.7554
──────────────────────────────────────────────────────


Hypotheses (crew):
H₀: All group means are equal (no effect of crew)
H₁: At least one group mean differs (effect of crew exists)
Test statistic: F(3, 76) = 4.906
p-value: 0.004
Conclusion: Reject the null hypothesis (p = 0.004 < α = 0.050)
```

- Applying the Kruskal-Wallis,

```
baltimare_harbor |> kruskal_wallis(outcome = unload_minutes,
                                   grouping = crew)
```

```
Kruskal-Wallis test

M[Anchor Crew] = 5.44 (n = 20)
M[Driftwood Crew] = 13.67 (n = 20)
M[Saltmarsh Crew] = 5.09 (n = 20)
M[Tidewater Crew] = 9.475 (n = 20)

Hypotheses:
H₀: All group distributions are equal (no effect of crew)
H₁: At least one group distribution differs in location (effect of crew exists)
Test statistic: H(3) = 7.917
p-value: 0.048
Conclusion: Reject the null hypothesis (p = 0.048 < α = 0.050)
```

- After choosing the appropriate method,
- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**